DIAGONAL QUADRATIC ENERGY CONTRACTION ON SECTION RETURNS OF THE CUBIC JACOBI–PERRON ALGORITHM

ARGUS OPEN RESEARCH PROJECT

Abstract. Whether every cubic algebraic vector has a periodic expansion under the Jacobi–Perron multidimensional continued fraction algorithm remains open, with $\mathbb{Q}(\sqrt[3]{D})$ for $D = 229$ being a classical test candidate. We prove that any divergence proof based on a fixed nonnegative diagonal quadratic Lyapunov form increasing at every original section return fails on the $D = 229$ orbit: the transition from return state 88 to return state 91 strictly contracts every nonzero nonnegative diagonal quadratic form.

1. Main Obstruction

Theorem 1.1. *Along the exact rational Jacobi–Perron orbit of* $(D^{1/3}, D^{2/3})$ *for* $D = 229$*, the section return from step* 88 *to step* 91 *satisfies* $|x_i(91)| < |x_i(88)|$ *in every coordinate, and therefore strictly decreases every quadratic form* $Q(x) = \sum_i w_i x_i^2$ *with* $w_i \geq 0$ *and* $\sum_i w_i > 0$*.*

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